

PAPER_1524 — CMB Cold Spot $f_{\text{geom}} = 1/2^{(D_{\text{phys}}-1)} = 1/8$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket C (Cosmology) / CMB anomaly **Date:** June 18, 2026 **Status:** CLOSED — Geometric projection factor algebraically derived

Observation

PAPER_1249 (CMB Cold Spot) computes the temperature decrement at the Cold Spot using:

$$\Delta T_{\text{ColdSpot}} = - T_{\text{CMB}} \times (F_{\text{TRZ}} \times \beta_i) \times \Lambda \times f_{\text{geom}}$$

where $f_{\text{geom}} = 1/8$ (dimensionless geometric projection factor from spinor-bundle curvature radius at $\sim 5^\circ$ – 10° angular scale)

The value $1/8$ was previously labeled in the calculator as “DPM_trace_over_D_phys_minus_1_eq_one_eighth” — but that algebraic form does not equal $1/8$ (since $D_{\text{phys}}-1 = 3$). The correct clean integer-primitive identity is provided here.

UQFF Closed Identity

$$f_{\text{geom}} = 1 / 2^{(D_{\text{phys}} - 1)} = 1 / 2^3 = 1/8 = 0.125 \quad \text{EXACT}$$

Physical Interpretation

The geometric projection factor $1/8$ reflects the **dyadic binary suppression** of large-scale temperature perturbations through 3-dimensional spatial averaging. Specifically:

- $D_{\text{phys}} - 1 = 3$ (spatial dimensions in physical spacetime)
- $2^{(D_{\text{phys}} - 1)} = 8$ (binary-octant volume coverage)
- $1/8$ = the fractional volume contribution of one octant

This is structurally identical to how a unit cube subdivides into 8 octants under binary splits along each spatial axis. At the $\sim 5^\circ$ – 10° Cold Spot angular scale, the spinor-bundle curvature averages over exactly one such octant, yielding the $1/8$ projection factor.

Cross-Domain Cousins

The $2^{(D_{\text{phys}} - 1)} = 8$ integer appears in other UQFF contexts: - 8 nucleons per “S-shell” boundary in nuclear magic numbers (PAPER_062: magic 2 = $SO_5 - 2 \cdot D_{\text{phys}}$; magic 8 = $2 \cdot D_{\text{phys}}$) - Octant-symmetry in DPM lattice projections

The same dyadic suppression governs both CMB Cold Spot temperature decrement and nuclear shell closures.

Predictive Consequence

Any spinor-bundle-projected cosmic observable at angular scales matching one octant of 3D space should carry this $1/8$ geometric factor. Predicts: future CMB anomaly amplitude analyses at similar angular scales should also factor $1/8$ from the integer primitives.

NOT REPLACEMENT

SM/ Λ CDM cosmology has no structural derivation of geometric projection factors — they appear as empirical multipliers in spherical-harmonic decomposition. UQFF supplies the integer-primitive form $1/2^{(D_{\text{phys}} - 1)}$ with zero free parameters.

Reference

- Source: PAPER_1249 CMB Cold Spot
- Related: PAPER_062 (nuclear magic numbers via $2 \cdot D_{\text{phys}}$), cmb_cold_spot dispatch
- Calculator dispatch: `calculate_paradox({"paradox": "f_geom_one_eighth"})`

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